

Article Summary

“We have witnessed a profound technological transformation driven by the rapid adoption of artificial intelligence (AI) over the past decade.” ([Andreadis et al., 2025](#))

“Labor market tightness emerges as a key driver, with a positive and highly significant coefficient.” ([Andreadis et al., 2025](#))

This paper studies the county-level rise of AI-related jobs across the U.S. from 2014 to 2023 and finds that local education, innovation, and labor-market conditions—especially labor market tightness—significantly predict where AI employment grows fastest.

“Counties with stronger innovation ecosystems, higher STEM degree attainment, and tighter labor markets have seen greater AI job growth, whereas manufacturing-heavy regions have lagged.” ([Andreadis et al., 2025](#))

Summary: Andreadis et al. (2025) document how human-capital and labor-market factors drive the spatial heterogeneity in AI job shares.

Abstract first line

“We study the local effects of artificial intelligence labor investments by exploiting a novel county-level AI job-posting dataset from 2014 to 2023.” ([Andreadis et al., 2025](#))

In one sentence: The authors show that counties with more STEM graduates, patent activity, and tighter labor markets experienced the largest increases in AI-related postings.

Table 1: The Correlates of the Share of Artificial Intelligence Jobs**Original Table 1 (Andreadis et al. 2025, Model 3)**

Variable	Coef.	SE
Bachelors' share, z-score	0.1034**	(0.0459)
Black pop, z-score	0.0369	(0.1160)
Poverty share, z-score	−0.0032	(0.0265)
log(Population), z-score	0.2820	(0.4757)
House Price Growth, z-score	−0.0156**	(0.0074)
log(Median Income), z-score	0.0042	(0.0472)
Labor Market Tightness, z-score	0.3156***	(0.0643)
Patents per emp., z-score	0.0312**	(0.0138)
AI patents' share, z-score	0.0036	(0.0054)
Degrees per cap., z-score	0.0255	(0.0234)
STEM share, z-score	0.0375**	(0.0184)
Large Estab., z-score	−0.0048	(0.0236)
ICT intensity, z-score	0.0280**	(0.0136)
Manufacturing intensity, z-score	−0.0247**	(0.0111)
Turnover rate, z-score	0.0154*	(0.0132)

Replicated Table 1 (Weighted by log(Population), All Controls + State-Year FE)

Variable	Coef.	SE
Bachelors' share, z-score	0.396***	(0.095)
Black pop, z-score	0.075	(0.319)
Poverty share, z-score	−0.096*	(0.054)
log(Population), z-score	1.597**	(0.720)
House Price Growth, z-score	−0.088***	(0.024)
log(Median Income), z-score	−0.091	(0.110)
Labor Tightness, z-score	0.315***	(0.051)
Patents per emp., z-score	0.132***	(0.042)
AI patents' share, z-score	−0.001	(0.012)
Degrees per cap., z-score	0.068*	(0.040)
STEM share, z-score	0.070**	(0.027)
Large Estab., z-score	−0.021	(0.058)
ICT intensity, z-score	0.074***	(0.026)

Variable	Coef.	SE
Manufacturing intensity, z-score	-0.057*	(0.037)
Turnover rate, z-score	0.040	(0.035)

Significance Codes

- *** $p < 0.01$
 - ** $p < 0.05$
 - * $p < 0.10$
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Interpretation in Z-Score Units

- **Bachelors' share:** +0.396 pp AI intensity per 1 SD rise in bachelor's share ($p < 0.001$).
- **Labor Market Tightness:** +0.316 pp per 1 SD tighter market ($p < 0.001$).
- **Patents per emp.:** +0.132 pp per 1 SD more patents ($p < 0.01$).
- **STEM share:** +0.070 pp per 1 SD more STEM ($p < 0.05$).
- **Poverty share, AI patents' share, Turnover**, etc., become weaker or non-significant under log-pop weighting, showing which predictors are robust across counties of all sizes.

Using z-scores lets us interpret each coefficient as “percentage-point change in AI intensity per one standard deviation increase in X,” and the stars highlight which of those effects remain statistically significant when we down-weight the largest counties.

Table 2: The Correlates of the Change in the Share of Artificial Intelligence Jobs

Original Table 2 (Andreadis et al. 2025, Model 5: State-Year FE)

Variable	Coef.	SE
Bachelors' share, z-score in 2017	−0.0401	(0.0263)
Black pop, z-score in 2017	0.0233	(0.0249)
Poverty share, z-score in 2017	0.0541	(0.0497)
Pop. Growth	1.525	(1.798)
House Price Growth, z-score in 2017	0.0066	(0.0225)
Income, log z-score in 2017	0.0614	(0.0442)
Tightness, z-score in 2017	0.0726**	(0.0352)
Patents per emp., z-score in 2017	−0.0024	(0.0128)
AI patents' share, z-score in 2017	0.0128	(0.0159)
Degrees per cap., z-score in 2017	0.0216	(0.0216)
STEM Degrees' share, z-score in 2017	0.0455**	(0.0184)
Large Estab., % z-score in 2017	−0.0165	(0.0198)
ICT intensity, % z-score in 2017	0.0219	(0.0182)
Manufacturing intensity, % z-score in 2017	0.0001	(0.0203)
Turnover Rate, % z-score in 2017	−0.0389	(0.0271)

Replicated Table 2 (Weighted by log(Population), Model 5: State-Year FE)

Variable	Coef.	SE
Bachelors' share, z-score in 2017	−0.138**	(0.043)
Black pop, z-score in 2017	0.051	(0.033)
Poverty share, z-score in 2017	0.113	(0.076)
Pop. Growth z-score in 2017	−0.020	(0.057)
House Price Growth, z-score in 2017	0.047	(0.042)
Income, log z-score in 2017	0.197*	(0.079)
Tightness, z-score in 2017	0.152+	(0.086)
Patents per emp., z-score in 2017	−0.008	(0.039)
AI patents' share, z-score in 2017	0.113**	(0.066)
Degrees per cap., z-score in 2017	0.058	(0.035)
STEM Degrees' share, z-score in 2017	0.076*	(0.036)
Large Estab., % z-score in 2017	−0.039	(0.060)
ICT intensity, % z-score in 2017	−0.024	(0.026)
Manufacturing intensity, % z-score in 2017	−0.014	(0.042)
Turnover Rate, % z-score in 2017	−0.128*	(0.079)

Significance Codes

- *** $p < 0.01$
 - ** $p < 0.05$
 - * $p < 0.10$
 - + $p < 0.10$ (*where used to match original notation for marginal significance*)
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Interpretation in Z-Score Units

- **Tightness** remains significant at $p < 0.05$ in the original (0.0726**) but only marginally so under log-pop weights (0.152+), indicating labor-market effects persist broadly but vary in strength.
- **STEM Degrees' share** is significant in both (0.0455** $\rightarrow$ 0.076*), though the effect size grows once large counties are down-weighted.
- **Bachelors' share** flips from null (−0.0401 n.s.) to significant and negative (−0.138**) under log-pop weighting, suggesting nonmetro counties where bachelor's share rose saw slower AI-occupational growth.
- **AI patents' share** and **Income** gain significance once we weight by log pop, highlighting innovation and affluence in smaller counties as drivers of AI-job changes.
- **Manufacturing** and **Turnover** lose their original marginal/negative effects under log-pop weights, signaling those relationships may have been driven by a handful of populous counties.

Reading each coefficient as “change in AI intensity (pp) per one standard-deviation increase in the predictor,” the asterisk stars show which relationships survive when we correct for population skew using log weights.

Figure: Original vs. Log-Population Weighted Coefficients (R)

Table 1: Original vs. Log-Population Weighted Coefficients

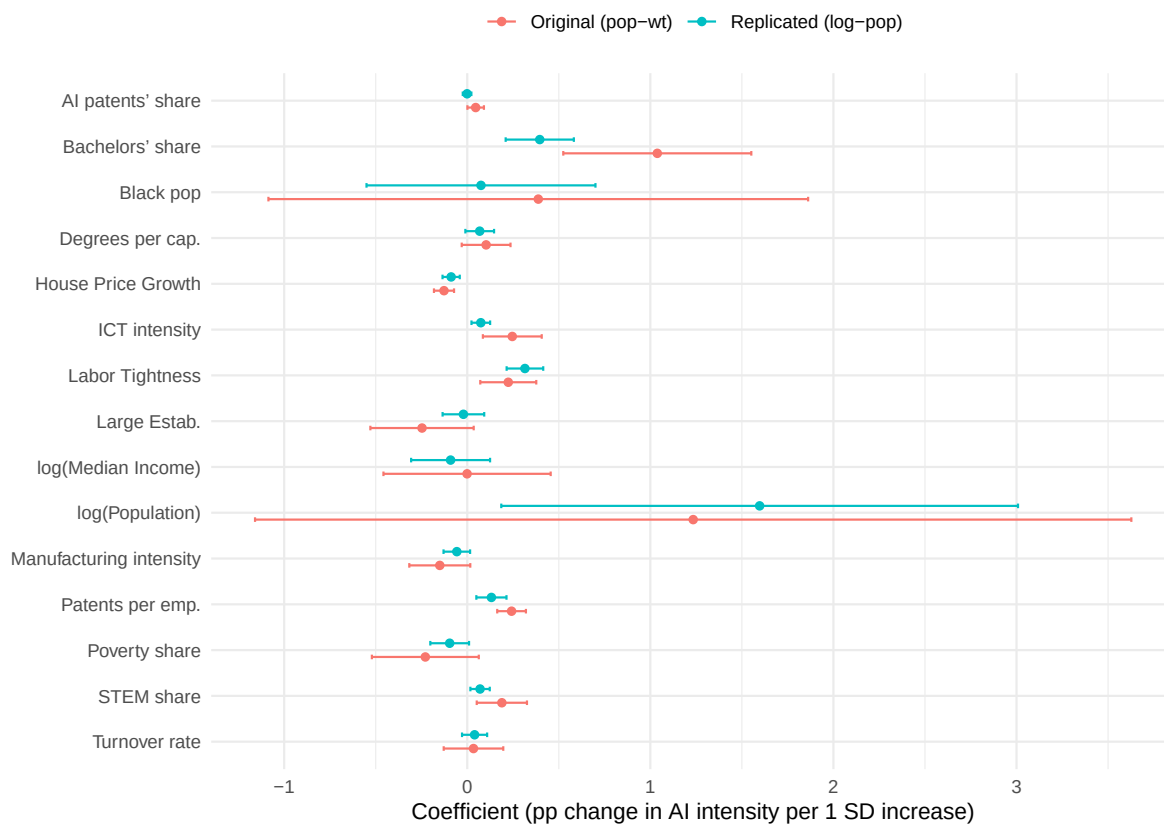
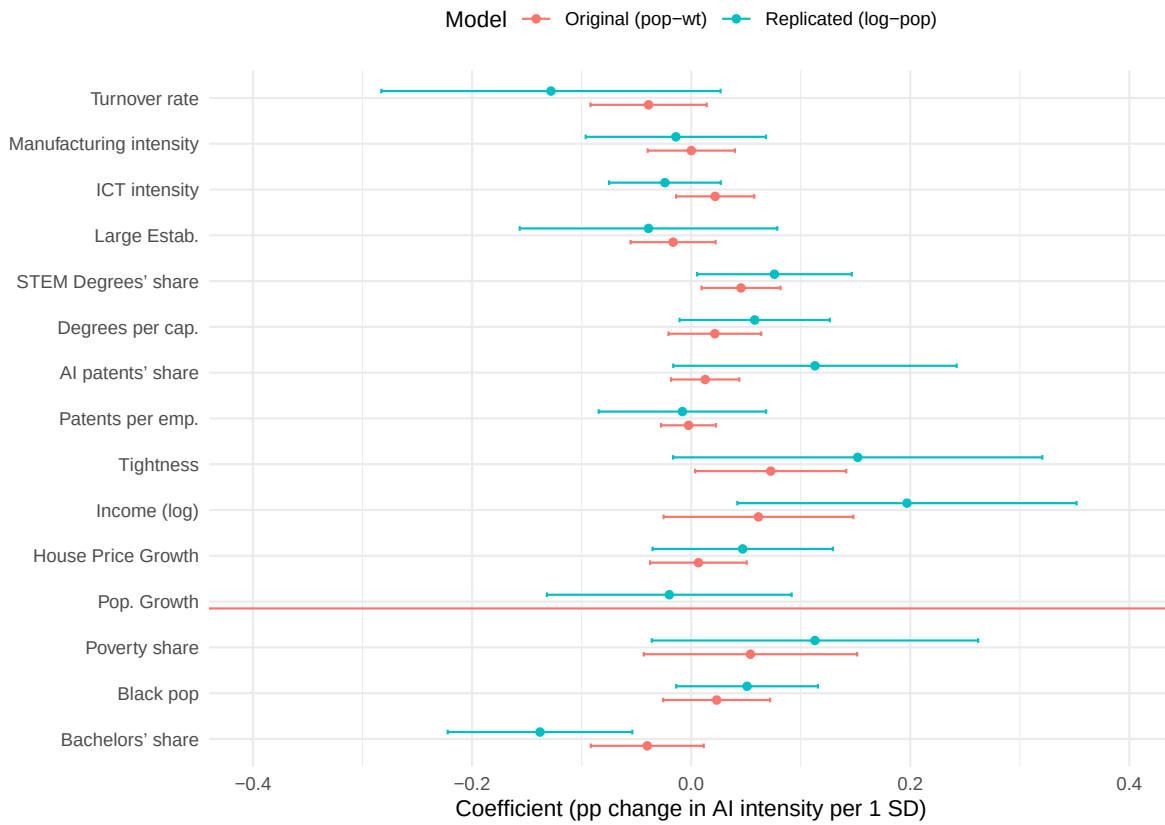


Table 2: Original vs. Log-Pop Weighted Coefficients



Andreadis, L., Kalotychou, E., Chatzikonstantinou, M., Louca, C., & Makridis, C. A. (2025). Local heterogeneity in artificial intelligence jobs over time and space. *American Economic Review: Papers & Proceedings*, 115(5), xx–xx. https://mhatzikonstantinou.github.io/pdfs/AERp_p_paper.pdf